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Task:

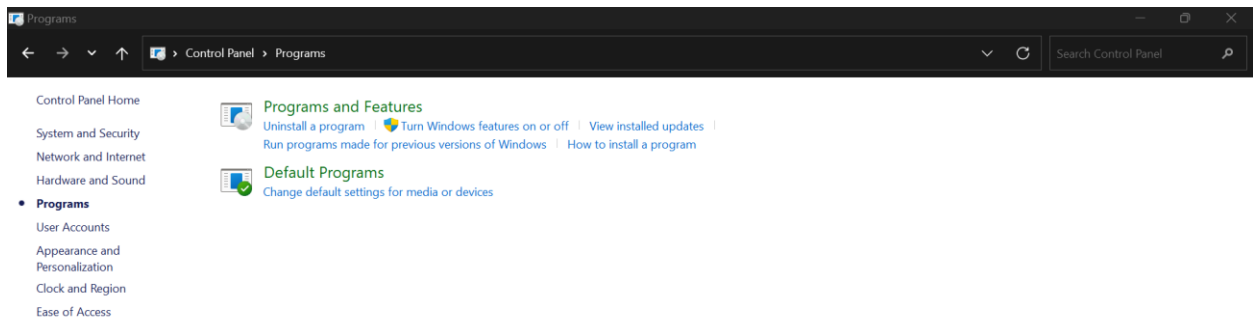
Create Python script that create share folder then copy data from desktop to remote server over the SMB share. Remote server SMB Share enabled username and password.

NOTE: Data Should be share only over the SMB Protocol.

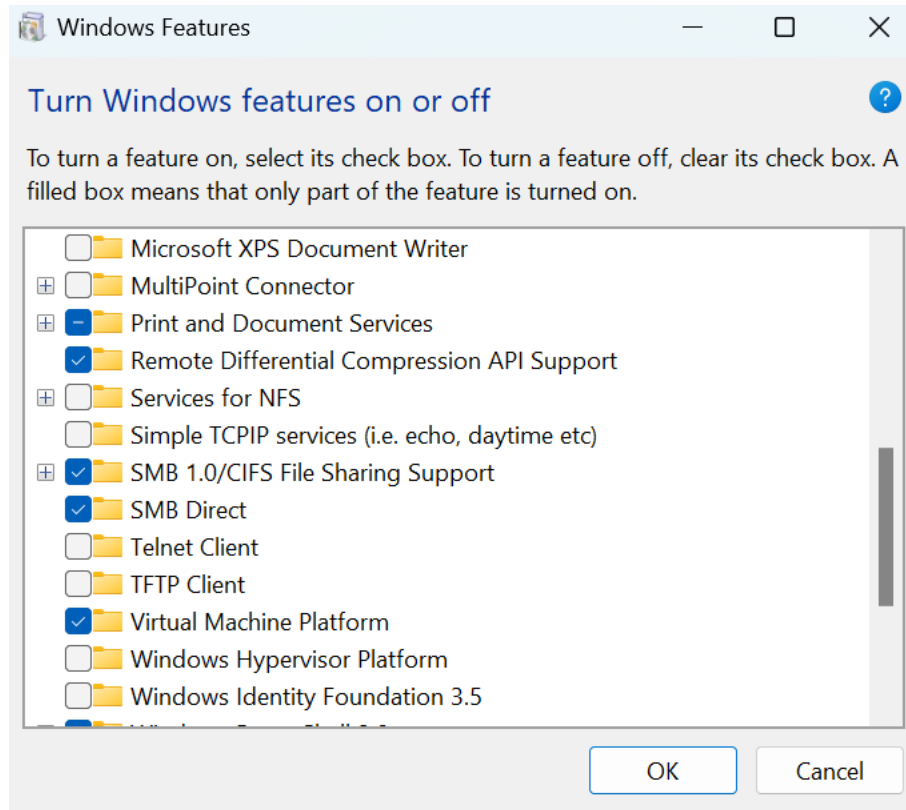
Setting Up an SMB Server on Windows

1. Enable SMB on Windows:

- Open **Control Panel**.
- Go to **Programs > Turn Windows features on or off**.

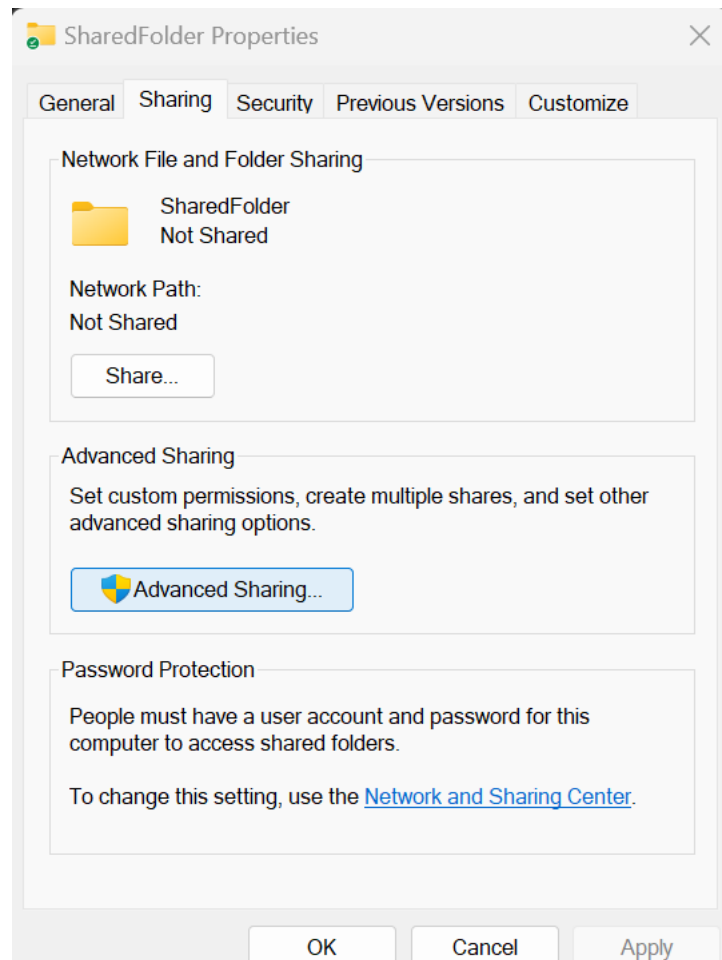


- Check **SMB 1.0/CIFS File Sharing Support** and **SMB Direct** (if available).
- Click **OK** and restart your computer if prompted.

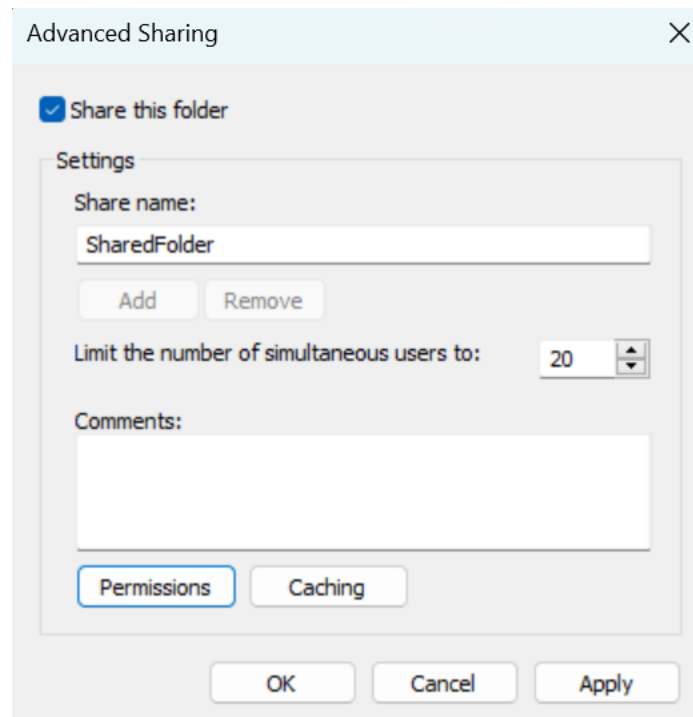


2. Create a Shared Folder:

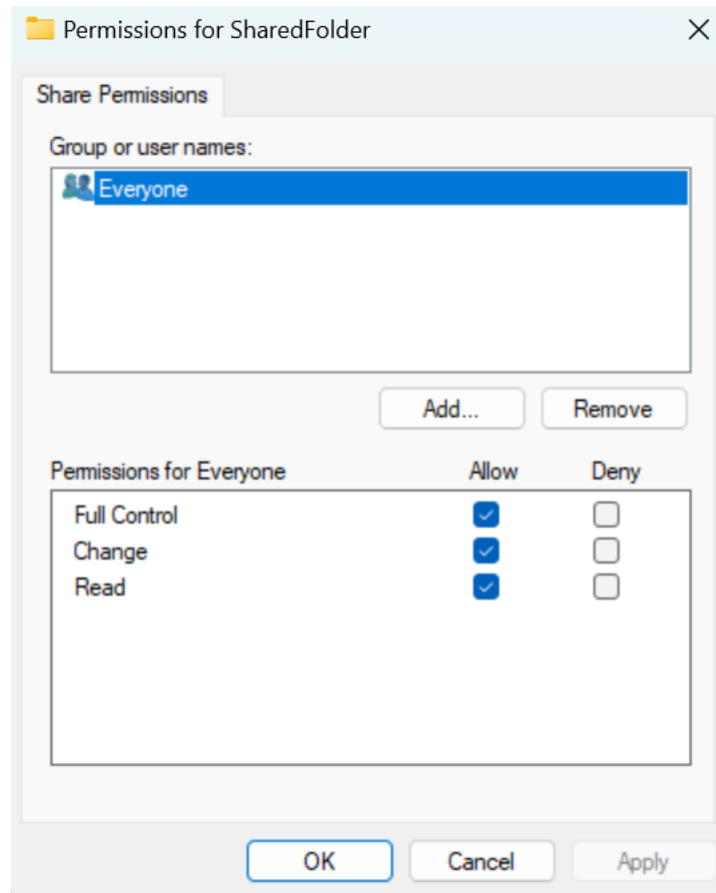
- Create a folder you want to share (e.g., C:\SharedFolder).
- Right-click the folder and select **Properties**.
- Go to the **Sharing** tab and click **Advanced Sharing**.



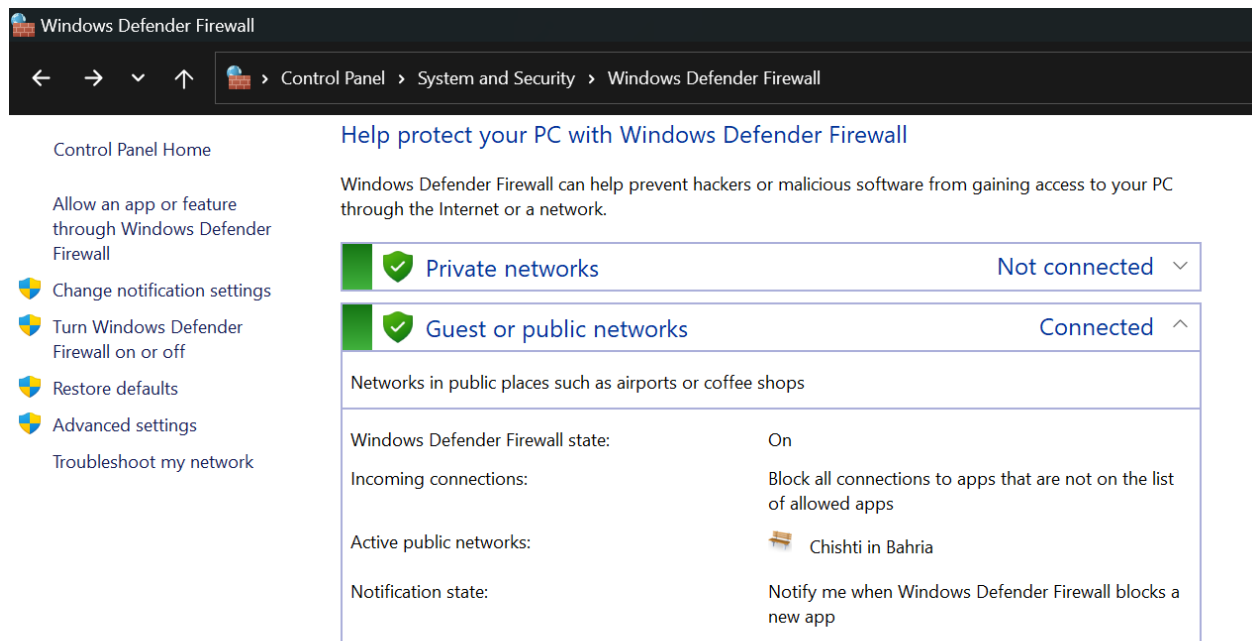
- Check **Share this folder**.
- Optionally, set the share name (default is the folder name).



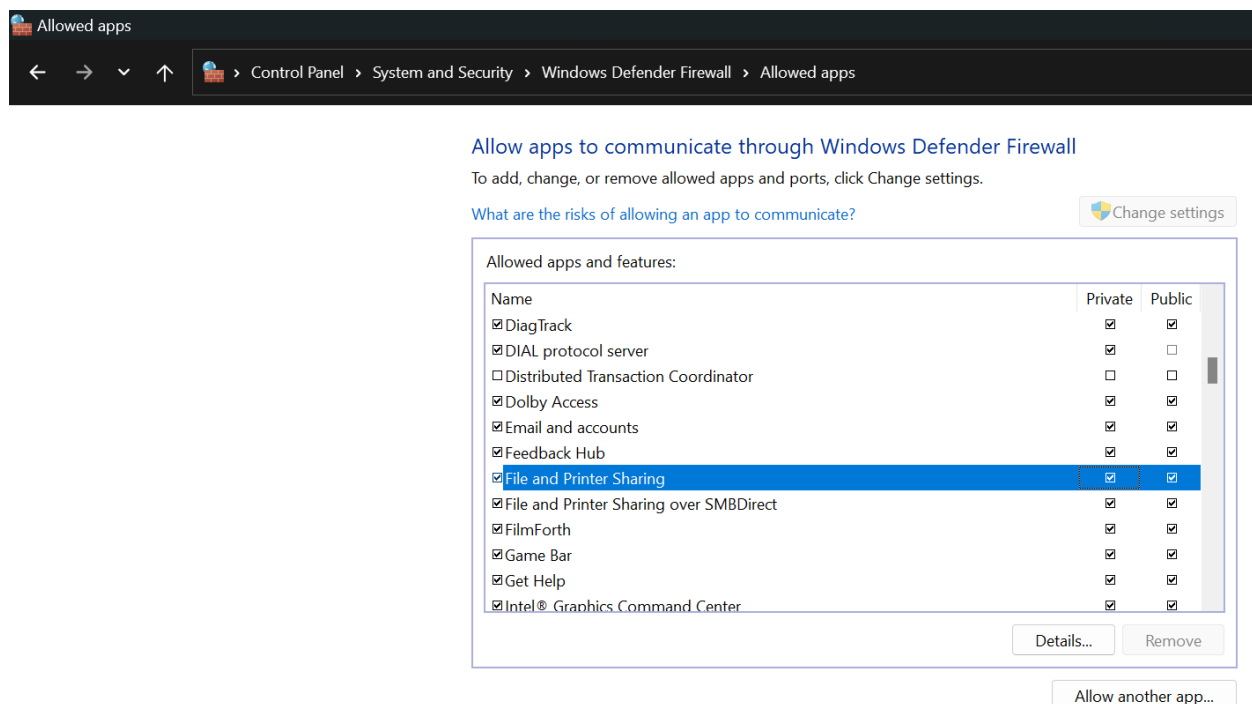
- Click **Permissions** and set the desired permissions (e.g., allow Full Control for specific users).



- Click **OK** to close the permissions and sharing settings.
3. **Configure Firewall (if needed):**
- Open **Control Panel > System and Security > Windows Defender Firewall**.
 - Click **Allow an app or feature through Windows Defender Firewall**.



- Ensure **File and Printer Sharing** is allowed on your desired network profile (private/public).



Creating a SAMBA server on parrot linux:

1. Setting Up the Remote SMB Server on Parrot Linux

Step 1: Install Samba on Parrot Linux

Samba is a software suite that allows file sharing between different operating systems, such as Windows and Linux, using the SMB/CIFS protocol.

Open a terminal on your Parrot Linux machine and install Samba:

```
sudo apt update  
sudo apt install samba
```

Step 2: Configure the Samba Server

1. **Create a Directory to Share:** Decide on a directory to share over the network. For example:

```
sudo mkdir -p /srv/samba/shared  
sudo chmod -R 0775 /srv/samba/shared  
sudo chown -R nobody:nogroup /srv/samba/shared
```

Edit the Samba Configuration File: Open the Samba configuration file in a text editor:

```
sudo nano /etc/samba/smb.conf
```

Add the following configuration at the bottom of the file:

```
[SharedFolder]  
path = /srv/samba/shared  
available = yes  
valid users = smbuser  
read only = no  
browsable = yes  
public = yes  
writable = yes
```

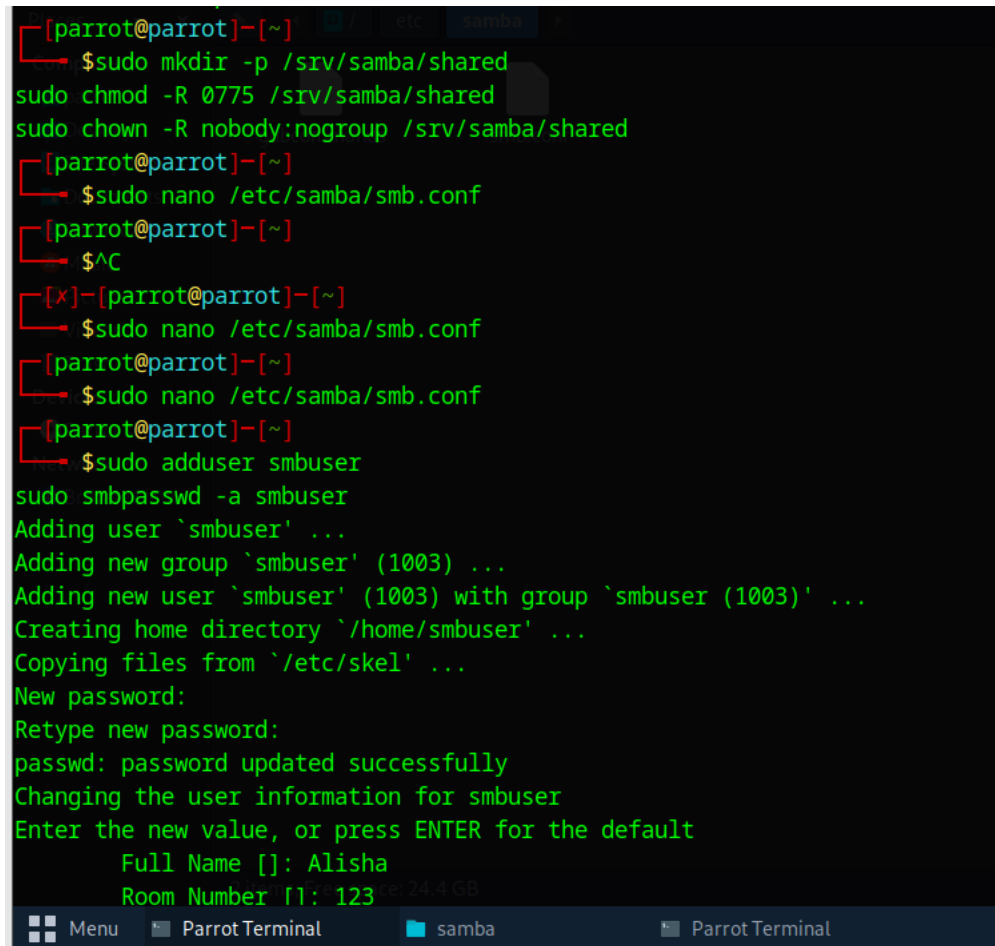
Replace smbuser with the username you will create in the next step.

2. **Create a Samba User:** Create a user for Samba. This user will be used to authenticate when connecting to the Samba server:

```
sudo adduser smbuser
```

```
sudo smbpasswd -a smbuser
```

You'll be prompted to create a password for this user. Remember this username and password, as you'll need them to connect from your Windows machine.



```
[parrot@parrot]~  
$sudo mkdir -p /srv/samba/shared  
sudo chmod -R 0775 /srv/samba/shared  
sudo chown -R nobody:nogroup /srv/samba/shared  
[parrot@parrot]~  
$sudo nano /etc/samba/smb.conf  
[parrot@parrot]~  
$^C  
[x]~[parrot@parrot]~  
$sudo nano /etc/samba/smb.conf  
[parrot@parrot]~  
$sudo nano /etc/samba/smb.conf  
[parrot@parrot]~  
$sudo adduser smbuser  
sudo smbpasswd -a smbuser  
Adding user `smbuser' ...  
Adding new group `smbuser' (1003) ...  
Adding new user `smbuser' (1003) with group `smbuser (1003)' ...  
Creating home directory `/home/smbuser' ...  
Copying files from `/etc/skel' ...  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for smbuser  
Enter the new value, or press ENTER for the default  
Full Name []: Alisha  
Room Number []: 123
```

3. **Restart the Samba Service:** After making changes, restart the Samba service to apply them:

```
sudo systemctl restart smbd
```

Step 3: Allow Samba Through the Firewall (if applicable)

If you have a firewall enabled on Parrot Linux, allow Samba traffic through it:

```
sudo ufw allow samba
```

```
[parrot@parrot]-[~]  
└─$ sudo systemctl restart smbd  
[parrot@parrot]-[~]  
└─$ sudo ufw allow samba  
Rules updated  
Rules updated (v6)  
[parrot@parrot]-[~]  
└─$
```

2 items, Free space: 24.4 GB

Menu Parrot Terminal samba Parrot Terminal

The credentials to put into the code:

```
Microsoft Windows [Version 10.0.22631.3810]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\ammok>hostname  
DESKTOP-V49PMKS  
  
C:\Users\ammok>|
```

Figure 1: Windows hostname

```
-[X]-[parrot@parrot]-[~]  
└─$ sudo ls -ld /srv/samba/shared  
drwxrwxr-x 1 smbuser smbuser 0 Aug 9 03:05 /srv/samba/shared  
└─[parrot@parrot]-[~]
```

```
Applications Places System Parrot Terminal
Browse and run installed applications
File Edit View Search Terminal Help

[parrot@parrot]~[~]user (1003) ...
$hostname: 'smbuser' (1003) with group 'smbuser (1003)' ...
parrot: home directory: '/home/smbuser' ...
[parrot@parrot]~[~]etc/skel' ...
$ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo mbuser
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1e:82:7a brd ff:ff:ff:ff:ff:ff
    altname enp2s1.123
    inet 192.168.58.129/24 brd 192.168.58.255 scope global dynamic noprefixroute ens33
        adding valid_lft 1696sec preferred_lft 1696sec to groups: 'users' ...
    inet6 fe80::538b:493a:e3b3:e0d/64 scope link noprefixroute
        New SMB valid_lft forever preferred_lft forever
[parrot@parrot]~[~] ...
$sudo pdbedit -L
[sudo] password for parrot:
smbuser:1003:Alisha,123,123,123,123
[parrot@parrot]~[~]
$sudo smbpasswd -a smbuser
New SMB password:
Retype new SMB password:
[parrot@parrot]~[~]
$sudo cat /etc/samba/smb.conf
#
```

Figure 2: On parrot Linux to check

```
Hostname
ip a
sudo pdbedit -L          #List Samba Users
sudo smbpasswd -a smbuser
sudo cat /etc/samba/smb.conf
```

Python Script on Windows

The Python script(smb_transfer.py) uses the pysmb library to connect to the SMB share and transfer files. Here's how the script works:

```
pip install pysmb
import os

from smb.SMBConnection import SMBConnection
```

```
# Path to the folder you want to share
```

```
local_directory = r"C:\Users\ammok\Desktop\Ascend Internship Tasks\Task#2-SMB_server\SharedFolder"
```

```
# Ensure the local directory exists
```

```
if not os.path.exists(local_directory):
```

```
    os.makedirs(local_directory)
```

```
# SMB server details
```

```
remote_server_name = 'parrot'
```

```
remote_server_ip = '192.168.58.129'
```

```
username = 'smbuser'
```

```
password = 'toor'
```

```
remote_share_name = 'SharedFolder'
```

```
local_machine_name = 'DESKTOP-V49PMKS'
```

```
# Create an SMB connection
```

```
conn = SMBConnection(username, password, local_machine_name, remote_server_name,  
use_ntlm_v2=True)
```

```
try:
```

```
    # Connect to the remote server
```

```
    assert conn.connect(remote_server_ip, 445)
```

```
# Specify the file to copy

source_file_path = os.path.join(local_directory, 'example.txt') # Replace with the actual file you
want to copy


# Ensure the file exists

if not os.path.isfile(source_file_path):

    raise FileNotFoundError(f"{source_file_path} does not exist")


# Open the file in binary mode for reading

with open(source_file_path, 'rb') as file_obj:

    # Copy the file to the remote server

    conn.storeFile(remote_share_name, 'example.txt', file_obj)

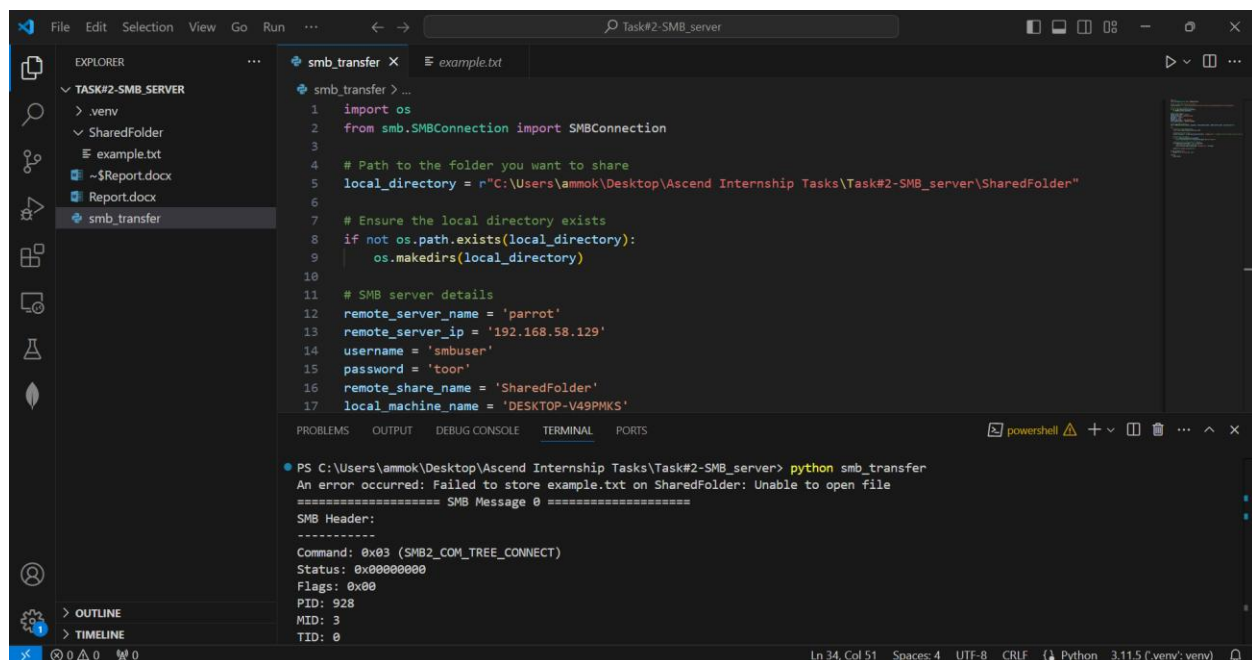
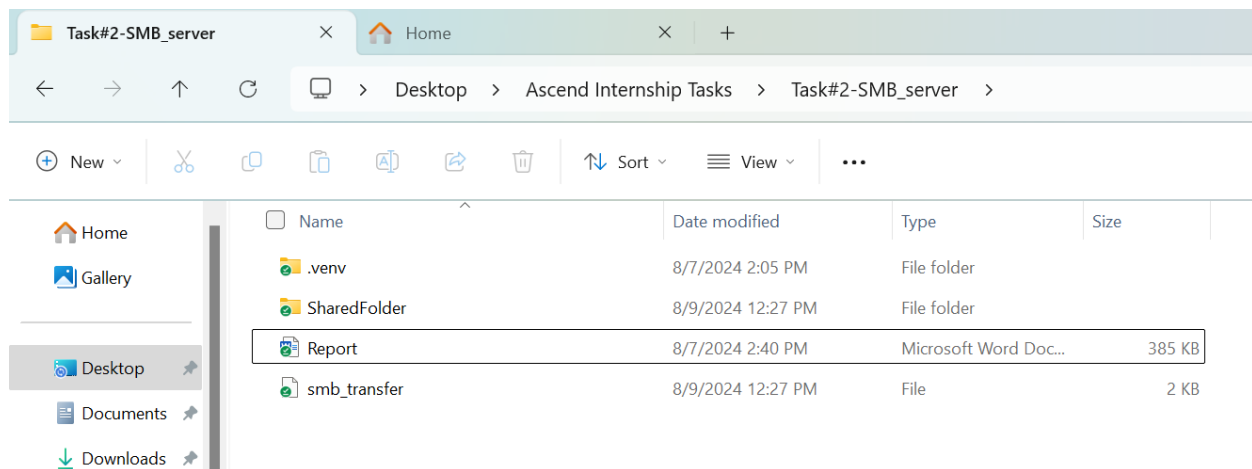

print("File copied successfully!")


except Exception as e:

    print(f"An error occurred: {e}")


finally:

    conn.close()
```



First run the script and then do this:

Testing the Connection:

- **Access the Shared Folder:**
 - In Windows File Explorer, enter \\192.168.58.129\SharedFolder in the address bar and press Enter.
 - Enter smbuser as the username and toor as the password if prompted.



Windows Security



Enter network credentials

Enter your credentials to connect to: 192.168.58.129

smbuser

☐

Remember my credentials

OK

Cancel